

Disruption of the Oral Microbiota Predicts Poor Outcomes after Allogeneic Hematopoietic Stem Cell Transplantation

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Background: The intestinal microbiota (IM) has been shown to play a vital role in regulating host immunity. In allogeneic hematologic stem cell transplantation (allo- HSCT), IM disruption also seems to affect the incidence of graft versus host disease (GVHD), overall survival (OS), and underlying disease relapse. The oral microbiota (OM) directly influences human health. Changes in the OM in patients undergoing allo- HSCT are known to be associated with respiratory signs and symptoms and oral mucositis; however, no correlation between OM and allo-HSCT outcomes have been reported to date.

Objectives: The study aims to evaluate whether the OM modulated the host systemic immune response and outcomes in patients who underwent allo-HSCT.

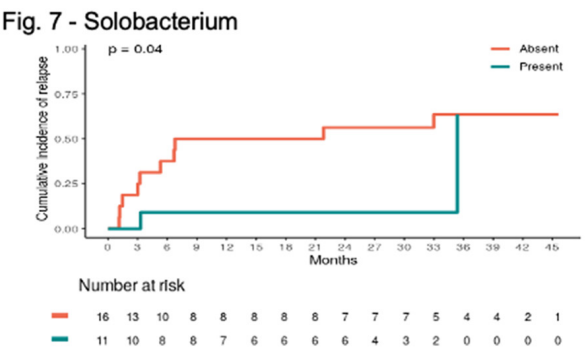
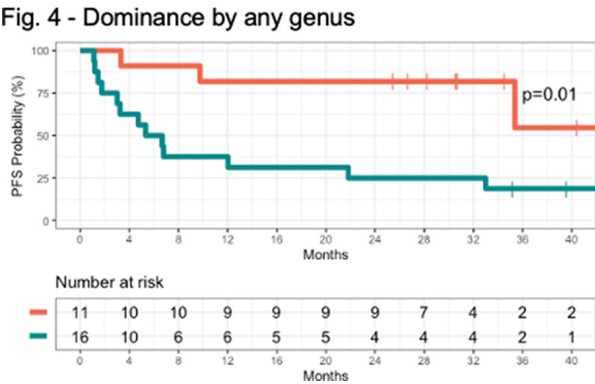
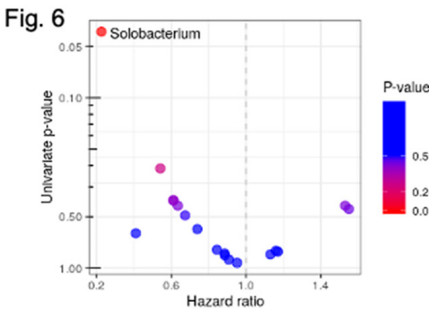
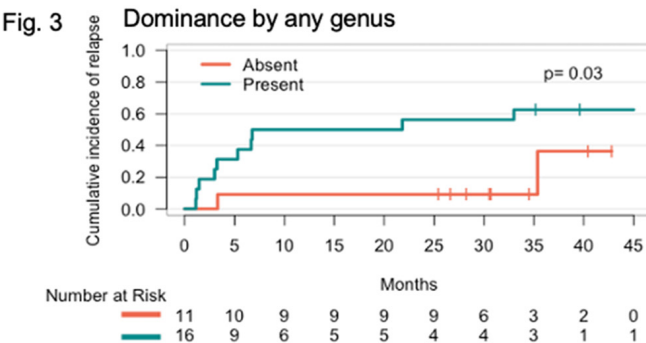
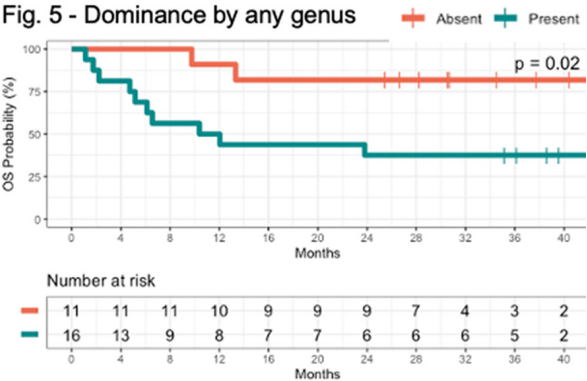
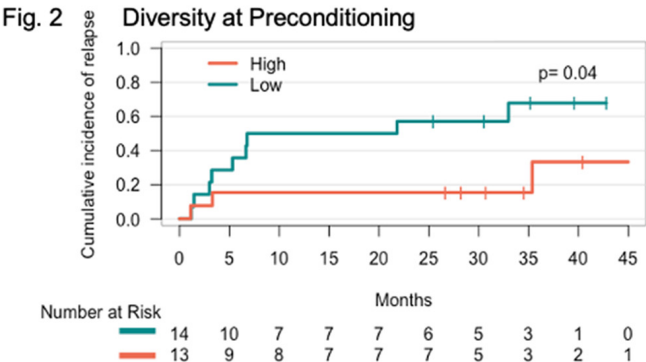
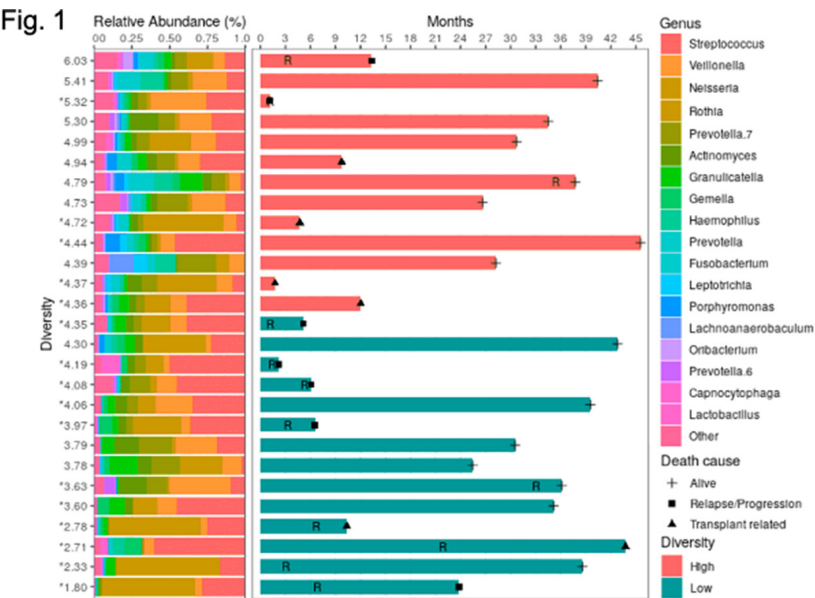
Patients and Methods: We collected samples from the oral mucosa of patients who underwent allo-HSCT between January 2016 and April 2018. Samples were collected at three time points: before the conditioning regimen, at aplasia, and at engraftment. *DNA extraction:* Bacterial cells were recovered from oral mucosa swabs through vortexing in TE buffer supplemented with 6 µL PureLink RNase A. DNA was extracted using a QIAamp DNA Blood Mini Kit and stored at -80°C. *16S rRNA amplicon sequencing:*

16S rRNA amplicon sequencing, 12.5ng DNA and prevalidated primers were used to amplify 16S rRNA hypervariable regions V3-V4. Amplicons were sequenced as described elsewhere on an Illumina MiSeq platform.

Results: We analyzed 30 patients. The most common diagnosis was acute leukemia (60%), most patients received reduced-intensity conditioning (60%). Donors were haploidentical in 33%, matched sibling in 30%, matched unrelated in 30%, and mismatched unrelated in 7%. Peripheral blood was the most common stem cell source (67%). We stratified patients into low or high diversity at each collection time. A swimmer plot was used to illustrate these correlations at preconditioning (Fig. 1). High OM diversity at preconditioning was associated with a lower risk of relapse at 3 years when compared with low diversity (33% versus 68%, respectively, $P = 0.04$; Fig. 2). The presence of dominance by any genus at preconditioning was also associated with an increased risk of relapse at 3 years when compared with the absence of dominance (63% versus 36%, respectively, $P = 0.03$; Fig. 3). The presence of dominance by any genus at preconditioning was also associated with inferior PFS (19% versus 55%, respectively, $P = 0.01$; Fig. 4) and OS (38% versus 81%, respectively, $P = 0.02$; Fig. 5). After adjusting analyses for the DRI, there was a trend toward a higher risk of relapse/progression in those with low OM diversity (HR = 0.30, 95% CI: 0.08-1.09, $P = 0.07$), and bacterial dominance of any genus remained significantly associated with the risk of relapse (HR = 4.19, 95% CI: 1.25-14.1, $P = 0.02$) and worse PFS (HR = 4.14, 95% CI: 1.15-14.89, $P = 0.03$); there was a tendency for bacterial dominance of any genus to also be related to worse OS (HR = 4.12, 95% CI: 0.89-19.13, $P = 0.07$). To further evaluate the association between preconditioning OM and transplant outcomes, we analyzed whether non-core genera were associated with a higher risk of relapse. Of the 18 genera that matched the selection criteria tested in univariate analysis for relapse (Fig. 6), only *Solobacterium* was significantly associated with lower relapse risk (9% versus 56%, respectively, $P = 0.04$; Fig. 7), and this association remained significant after adjusting for DRI (HR = 0.20, 95% CI: 0.06-0.67, $P = 0.01$).

Conclusion: High OM diversity and presence of dominance by any genus (more importantly *Solobacterium*) at preconditioning was associated with a lower risk of relapse. Prospective trials and validation cohorts are needed to confirm these findings and to test whether early interventions to correct OM dysbiosis or more aggressive strategies to prevent relapse in OM dysbiotic patients, such as early

immunosuppression withdrawal, maintenance therapy, or prophylactic donor lymphocyte infusions, could improve allo-HSCT outcomes.



Disclosures

No relevant conflicts of interest to declare.

Author notes

* Asterisk with author names denotes non-ASH members.

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